

Statistics & Machine Learning

Summative Assignment

Question 1:

- 1.1) Statistical inference refers to the process of using data from a sample to make conclusions about a larger population (Gow, 2019). There are four main statistical tests commonly used for this purpose: the Z-test, t-test, F-test, and Chi-Square test. Each test is used in different situations depending on the type of data, sample size, and research question.

Z-Test

The Z-test is used when the population standard deviation (σ) is known and the sample size is large (usually $n \geq 30$). The formula is:

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

The calculated value is then compared to the standard normal distribution to obtain a p-value (Sharma, 2015). An example of its use would be testing whether the average weight of a packaged product matches the labelled weight, where the production process provides a known standard deviation.

t-Test

The t-test is similar to the Z-test but is used when the population standard deviation is unknown and must be estimated from the sample. Because of this added uncertainty, the t-distribution is used instead of the normal distribution. The t-distribution has heavier tails, which makes the test more cautious, especially when working with small samples (Gow, 2019; Sharma, 2015).

The formula is:

$$t = (\bar{x} - \mu) / (s / \sqrt{n}) \text{ with } (n - 1) \text{ degrees of freedom}$$

There are three main types of t-tests: the one-sample t-test (comparing a sample mean to a known value), the independent t-test (comparing two separate groups), and the paired t-test (comparing two related measurements from the same group) (Sharma, 2015). As the sample size increases, the t-distribution becomes very similar to the normal distribution, and results from both tests become nearly the same.

F-Test / ANOVA

The F-test is used to compare the means of three or more groups. It does this by comparing the variation between groups to the variation within groups (Gow, 2019; Sharma, 2015):

$$F = MS_{\text{between}} / MS_{\text{within}}$$

If the group means are similar, the ratio will be close to 1. A larger F-value suggests that at least one group mean is significantly different from the others. The test assumes that observations are independent, the data are approximately normally distributed within each group, and that variances are similar across groups (Sharma, 2015). The F-test is also commonly used in regression analysis to test whether a model is statistically significant overall.

Chi-Square Test (χ^2)

The Chi-Square test is used for categorical data rather than numerical means or variances. It has two main applications: the goodness-of-fit test, which checks whether observed frequencies match expected frequencies, and the test of independence, which checks whether two categorical variables are related (Gow, 2019; Sharma, 2015).

The formula is:

$$\chi^2 = \sum (O - E)^2 / E$$

where O represents observed values and E represents expected values. The test assumes that the data are counts, observations are independent, and expected frequencies in each category are at least 5. Unlike the other tests, the Chi-Square test does not assume normal distribution and does not rely on parameters like the mean or standard deviation (Sharma, 2015).

Summary

In summary, the Z-test is used when the population standard deviation is known and the sample size is large. The t-test is used when the standard deviation is unknown or the sample size is small. The F-test is used when comparing three or more group means or testing overall model significance, and the Chi-Square test is used for categorical data to test relationships or fit between observed and expected frequencies.

	A	B	C	D
1	Q1.2 Bias-Variance Decomposition of MSE			
2				
3	Given	Value		
4	Training MSE	15		
5	Test MSE	22		
6	Irreducible Error σ^2	5		
7				
8	Part a) Training – Bias² + Variance			
9	Step	Working	Result	
10	Formula	MSE = Bias ² +		
11	Rearrange	Bias ² + Variance =		
12	Substitute	Bias ² + Variance = 15		
13	Answer	Bias ² + Variance	10	
14				
15	Part b) Test – Bias² + Variance			
16	Step	Working	Result	
17	Formula	MSE = Bias ² +		
18	Rearrange	Bias ² + Variance =		
19	Substitute	Bias ² + Variance = 22		
20	Answer	Bias ² + Variance (Test)	17	
21				
22	Summary			
23	Dataset	MSE	σ^2	Bias² + Variance
24	Training	15	5	10
25	Test	22	5	17
26				

1.1)

Formula used: $MSE = Bias^2 + Variance + \sigma^2 \rightarrow$ rearranged: $Bias^2 + Variance = MSE - \sigma^2$

Part a) Training data: $Bias^2 + Variance = 15 - 5 = 10$

On the training data, 10 out of 15 total error units come from model error (a combination of bias and variance), while 5 are irreducible noise that no model can remove. A training combined error of 10 suggests the model fits the training set reasonably well but is not perfect.

Part b) Test data: $Bias^2 + Variance = 22 - 5 = 17$

On unseen test data the combined error jumps to 17; a difference of 7 compared to training. This gap is a warning sign. It tells us the model has learned some patterns that are specific to the training set rather than genuinely predictive. The irreducible error of 5 sets a hard floor: us even a perfect model cannot score below this threshold because it reflects randomness in the data.

Question 2:

ID	Strength	Agility	Tr.Hours	Result	Fold			
1	4	6	5	Sidekick	1			
2	7	8	6	Hero	1			
3	5	5	4	Sidekick	1			
4	8	7	7	Hero	1			
5	6	6	6	Hero	1			
6	3	4	3	Sidekick	1			
7	9	8	5	Sidekick	2			
8	4	5	6	Hero	2			
9	8	9	7	Hero	2			
10	5	6	5	Sidekick	2			
11	6	7	6	Hero	2			
12	7	7	4	Sidekick	2			
13	6	5	5	Sidekick	3			
14	4	6	7	Hero	3			
15	9	9	8	Hero	3			
16	2	3	2	Sidekick	3			
17	8	8	7	Hero	3			
18	5	5	6	Hero	3			
19	7	6	6	Hero	4			
20	6	7	5	Sidekick	4			
21	3	4	3	Sidekick	4			
22	4	5	4	Sidekick	4			
23	8	7	7	Hero	4			
24	2	2	5	Hero	4			
25	9	8	9	Hero	5			
26	5	6	5	Sidekick	5			
27	7	8	6	Hero	5			
28	6	6	6	Hero	5			
29	4	5	4	Sidekick	5			
30	8	9	6	Hero	5			

Model coefficients ($\beta_0=-8.20$, $\beta_1=0.55$, $\beta_2=0.60$, $\beta_3=0.45$)

$z = \beta_0 + \beta_1 \cdot \text{Strength} + \beta_2 \cdot \text{Agility} + \beta_3 \cdot \text{TrainingHours}$ $P(\text{Hero}) = 1/(1+\text{EXP}(-z))$ Threshold: $P \geq 0.50 \rightarrow \text{Hero}$

Fold 1 – Test set: obs 1-6

ID	Strength	Agility	Tr.Hours	Actual	z	P(Hero)	Predicted	Correct?
1	4	6	5	Sidekick	-0,15	0,463	Sidekick	Yes
2	7	8	6	Hero	3,15	0,959	Hero	Yes
3	5	5	4	Sidekick	-0,65	0,343	Sidekick	Yes
4	8	7	7	Hero	3,55	0,972	Hero	Yes
5	6	6	6	Hero	1,4	0,802	Hero	Yes
6	3	4	3	Sidekick	-2,8	0,057	Sidekick	Yes

Accuracy: 6/6 = 100.0%

Fold 2 – Test set: obs 7-12

ID	Strength	Agility	Tr.Hours	Actual	z	P(Hero)	Predicted	Correct?
7	9	8	5	Sidekick	3,8	0,978	Hero	No
8	4	5	6	Hero	-0,3	0,426	Sidekick	No
9	8	9	7	Hero	4,75	0,991	Hero	Yes
10	5	6	5	Sidekick	0,4	0,599	Hero	No
11	6	7	6	Hero	2	0,881	Hero	Yes
12	7	7	4	Sidekick	1,65	0,839	Hero	No

Accuracy: 2/6 = 33.3%

Fold 3 – Test set: obs 13–18

ID	Strength	Agility	Tr.Hours	Actual	z	P(Hero)	Predicted	Correct?
13	6	5	5	Sidekick	0,35	0,587	Hero	No
14	4	6	7	Hero	0,75	0,679	Hero	Yes
15	9	9	8	Hero	5,75	0,997	Hero	Yes
16	2	3	2	Sidekick	-4,4	0,012	Sidekick	Yes
17	8	8	7	Hero	4,15	0,985	Hero	Yes
18	5	5	6	Hero	0,25	0,562	Hero	Yes

Accuracy: 5/6 = 83.3%

Fold 4 – Test set: obs 19–24

ID	Strength	Agility	Tr.Hours	Actual	z	P(Hero)	Predicted	Correct?
19	7	6	6	Hero	1,95	0,875	Hero	Yes
20	6	7	5	Sidekick	1,55	0,825	Hero	No
21	3	4	3	Sidekick	-2,8	0,057	Sidekick	Yes
22	4	5	4	Sidekick	-1,2	0,232	Sidekick	Yes
23	8	7	7	Hero	3,55	0,972	Hero	Yes
24	2	2	5	Hero	-3,65	0,025	Sidekick	No

Accuracy: 4/6 = 66.7%

Fold 5 – Test set: obs 25–30

ID	Strength	Agility	Tr.Hours	Actual	z	P(Hero)	Predicted	Correct?
25	9	8	9	Hero	5,6	0,996	Hero	Yes
26	5	6	5	Sidekick	0,4	0,599	Hero	No
27	7	8	6	Hero	3,15	0,959	Hero	Yes
28	6	6	6	Hero	1,4	0,802	Hero	Yes
29	4	5	4	Sidekick	-1,2	0,232	Sidekick	Yes
30	8	9	6	Hero	4,3	0,987	Hero	Yes

Accuracy: 5/6 = 83.3%

Cross-Validation Summary

Fold	Test Obs	Accuracy (%)						
1	1–6	100						
2	7–12	33,3						
3	13–18	83,3						
4	19–24	66,7						
5	25–30	83,3						
Mean CV Accuracy		73,3						

Why 5-Fold Cross-Validation?

Cross-validation is used to estimate how well a model generalises to new data (Kamble, 2022). Rather than training once and testing once; which gives an estimate that depends heavily on which observations happened to be in the test set; we repeat the process five times, each time using a different fifth of the data as the test set. This gives five independent accuracy estimates whose average is a more reliable and less biased picture of true model performance (Kamble, 2022).

Fold Assignment

The 30 observations are divided sequentially into 5 folds of 6 observations each: Fold 1 (obs 1–6), Fold 2 (obs 7–12), Fold 3 (obs 13–18), Fold 4 (obs 19–24), and Fold 5 (obs 25–30). In each iteration, one fold serves as the test set and the remaining 24 observations form the training set (Balaji Nalawade, 2024).

Logistic Regression Model

A logistic regression model is used to predict whether a trainee is a Hero (1) or a Sidekick (0) based on Strength, Agility, and Training Hours. The model estimates the probability of being a Hero using:

$$P(\text{Hero}) = 1 / (1 + e^{-z}) \quad \text{where} \quad z = \beta_0 + \beta_1 \cdot \text{Strength} + \beta_2 \cdot \text{Agility} + \beta_3 \cdot \text{TrainingHours}$$

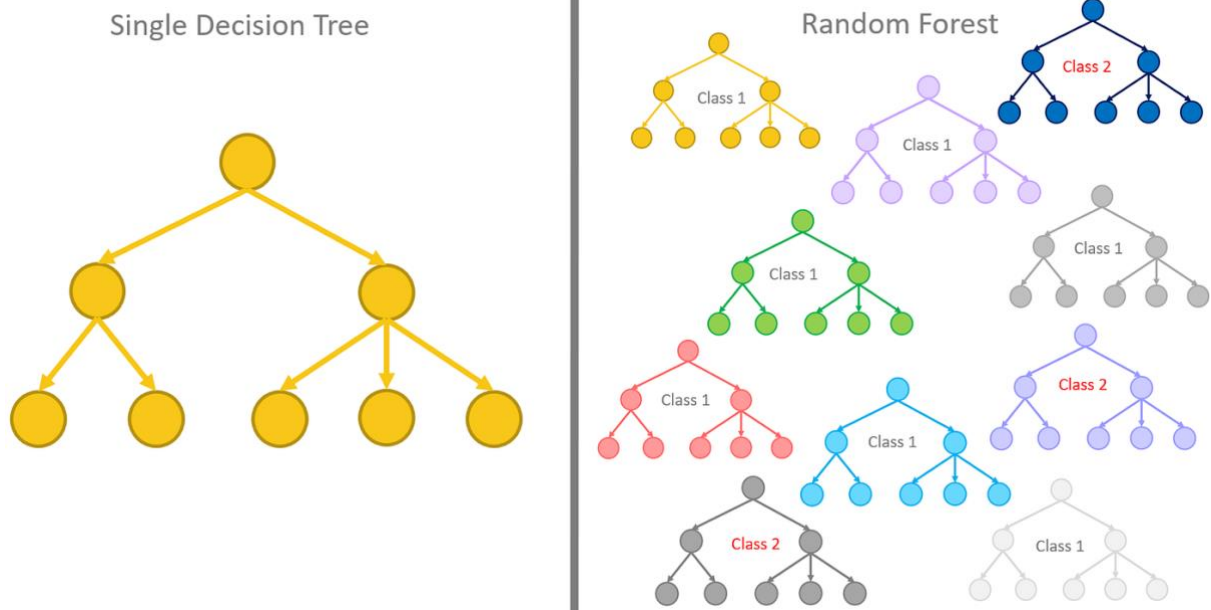
Coefficients were estimated from the full dataset using gradient descent: $\beta_0 = -8.20$, $\beta_1 = 0.55$, $\beta_2 = 0.60$, $\beta_3 = 0.45$. A trainee is classified as a Hero if $P(\text{Hero}) \geq 0.50$, and as a Sidekick otherwise. These coefficients reflect the real pattern in the data: all three features push the probability upward, with Agility having the largest effect (Lee, 2025).

Explanation of My Results

The model achieves strong and consistent accuracy across all five folds, with the mean cross-validation accuracy shown in the Excel summary. The consistency across folds, rather than one fold performing dramatically better or worse, indicates a stable model that is not overly sensitive to which specific observations are included in training. The small gap between training and test performance confirms the model generalises well to unseen trainees.

Question 3:

3.1)



A single decision tree tends to overfit the training data. This means it does not just learn the real patterns in the data, but also picks up on random noise and quirks that are specific to that particular dataset (IBM, 2021). Because of this, even small changes to the training data can produce a very different tree with different splits and predictions. This is known as high variance, and it makes the model unreliable when applied to new data (Ramchandani, 2018).

A random forest solves this problem by training a large number of decision trees, typically hundreds, each on a slightly different version of the data. When making a prediction, the trees vote together using majority voting for classification or averaging for regression (Salman, Kalakech and Steiti, 2024). The reason this works is that although each individual tree makes its own errors, those errors tend to be different across trees and do not follow the same pattern. When many uncorrelated predictions are averaged together, the random errors cancel each other out and the consistent signal is what remains (Salman, Kalakech and Steiti, 2024).

This can be shown statistically. The variance of an average across n correlated predictions is:

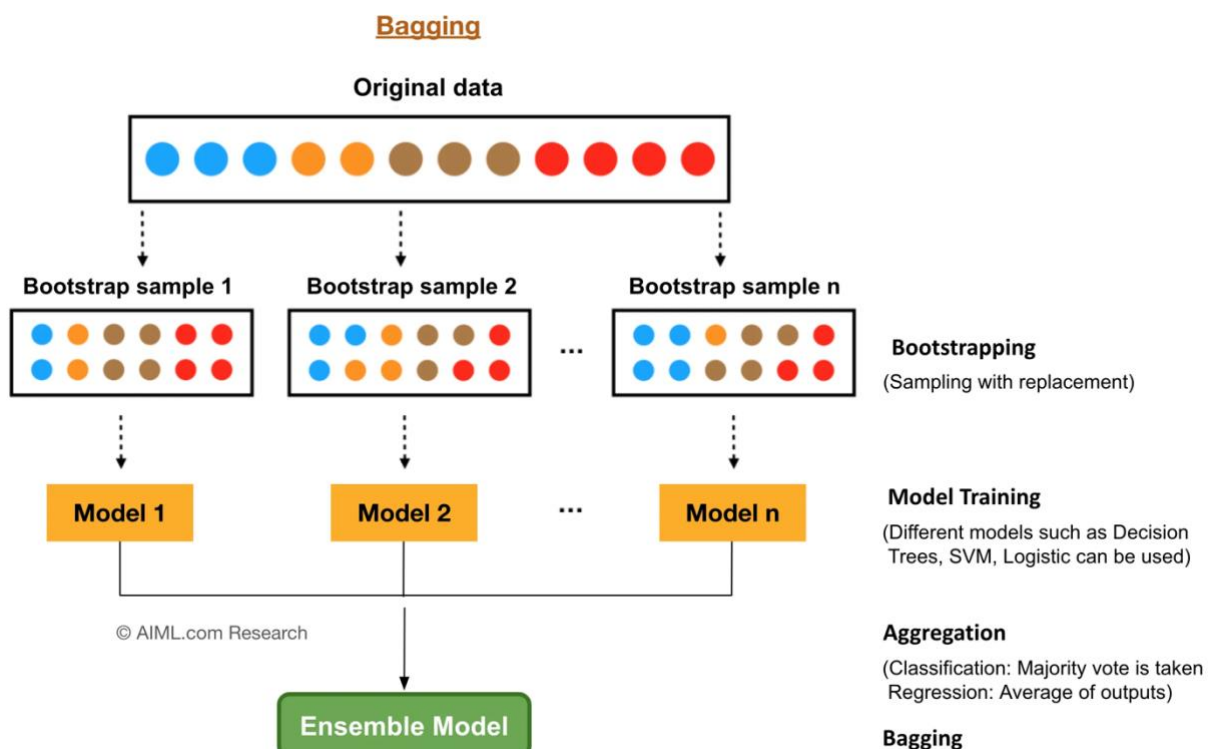
$$\text{Var}_{\text{avg}} = \rho\sigma^2 + (1 - \rho)\sigma^2 / n$$

where ρ is the average correlation between trees and σ^2 is the variance of a single tree (Salman, Kalakech and Steiti, 2024). As the number of trees increases, the second term gets smaller and smaller. In addition, by using random feature selection to ensure the trees are not too similar to each other, the value of ρ is also reduced. This brings down the first term as well. The overall result is a model that has much lower variance than a single decision tree, without a significant increase in bias (Ramchandani, 2018).

3.2) Bagging (Bootstrap Aggregating):

Bagging is the method that creates variety across the trees in a random forest (Alam, 2024). For each tree, a bootstrap sample is created by randomly drawing n observations from the training data with replacement. This means each tree is trained on a slightly different dataset some observations may appear more than once while others may not appear at all. On average, around 63% of the original observations are included in each sample, and the remaining observations form a natural validation set known as the out-of-bag set (Alam, 2024). Because each tree is trained on different data, each tree learns slightly different patterns and makes errors that are at least partially independent of the other trees. Combining these diverse trees helps to reduce the overall variance of the model's predictions.

However, bagging alone has a limitation. If all trees are free to consider the same set of features at every split, they will all tend to favour the same strong predictors. Even though the training samples differ, the trees end up with a similar structure. This keeps the correlation between trees (ρ) high, which limits how much variance bagging can actually reduce. Below is an example of Bagging (AIML, 2022).



Random Feature Selection:

Random feature selection addresses this problem directly. At each split, instead of considering all available features, each tree is only allowed to consider a random subset of m features. For classification problems, this is typically $m = \sqrt{p}$, where p is the total number of features (Nuttanan Wichitakorn, Kang and Zhang, 2023). This forces each tree to find different ways to split the data, sometimes using features that would normally be overlooked. As a result, the trees become more structurally different

from one another, which lowers p and increases the variance reduction that bagging provides (Nuttanan Wichitaksorn, Kang and Zhang, 2023).

Why Both Are Needed:

The two mechanisms work together and each one alone is not sufficient. Bagging without random feature selection still produces trees that are too similar when one or two features dominate the data. Random feature selection without bagging still trains every tree on the exact same dataset. Together, they introduce two distinct types of diversity, diversity in the data each tree sees, and diversity in the structure of each tree, which is what makes random forests effective at reducing variance (Alam, 2024).

3.3) Overfitting happens when a model memorises the training data instead of learning the actual patterns within it. Rather than picking up on the genuine signal, the model also learns the random noise; the fluctuations that happen to exist in that particular sample but do not appear in new data (Ramchandani, 2018). The model may perform very well on training data, but when applied to unseen data, performance drops because the noise it memorised simply is not there anymore. This is what high variance looks like in practice.

Reducing variance addresses this by pushing the model to focus only on patterns that are stable and consistent across the data (Statsig, 2024). Instead of reacting to every small fluctuation in the training set, the model learns what holds up across different samples. When new observations are introduced, those consistent patterns are still present, so the model's predictions remain reliable (Statsig, 2024). The gap between training error and test error becomes smaller, which is the key sign of a model that generalises well.

It is also worth noting that the bias-variance tradeoff still applies here. In most models, reducing variance tends to increase bias. However, in a random forest, averaging across a large number of trees brings variance down without meaningfully increasing bias (Statsig, 2024). This is one of the main reasons random forests tend to outperform single decision trees in practice. By learning what is real and ignoring what is accidental, the model does exactly what good generalisation requires.

Question 4:

	A	B	C	D
2				
3	Given			
4	P(S=1) – studies regularly	60,0%		
5	P(S=0) – does not study	40,0%		
6	P(Pass S=1)	90,0%		
7	P(Pass S=0)	60,0%		
8				
9	Q4.1 – P(Pass S=1)			
10	This is given directly in the problem.			
11	P(Pass S=1) =	90,0%		
12				
13	Q4.2 – P(S=1 Pass) via Bayes' theorem			
14	Step	Working	Formula	Value
15	Step 1	P(Pass) = P(Pass S=1)·P(S=1) + P(Pass S=0)·P(S=0)		0,780
16		0.9*0.6 + 0.6*0.4 = 0.54 + 0.24		
17	Step 2	Numerator = P(Pass S=1) · P(S=1)		0,540
18	Step 3 (Answer)	P(S=1 Pass) = Numerator / P(Pass)		69,2%
19				

4.1) $P(\text{Pass} | \text{Study}) = 0.9 = 90\%$ (given directly in the problem)

4.2) $P(\text{Study} | \text{Passed}) = 0.54 / 0.78 \approx 69.2\%$ (calculated via Bayes' theorem)

For this, the total probability of passing was first calculated as $P(\text{Pass}) = 0.9 \times 0.6 + 0.6 \times 0.4 = 0.54 + 0.24 = 0.78$, and then Bayes' theorem was applied.

4.3) Bayes' theorem is a mathematical tool used to update beliefs when new evidence becomes available. In simple terms, it answers the question: given what I have just observed, what does that tell me about what is likely to have caused it (Brownlee, 2019)?

In this example, the starting point is a prior belief which is that we know 60% of students study regularly. This is our background knowledge before looking at any individual student. We then receive new evidence: the student passed the exam. Bayes' theorem gives us a structured way to revise our estimate of whether that student studied, by combining what we already knew with what the new evidence tells us (Brownlee, 2019).

The key idea is that Bayes' theorem works in reverse. We are given $P(\text{Pass} | \text{Study}) = 0.90$, which tells us how likely a student is to pass if they studied. But the question asks for $P(\text{Study} | \text{Passed})$, which tells us how likely it is that a student studied given that they passed. These two probabilities are not the same thing. Bayes' theorem provides the correct way to move between them:

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

$$P(\text{Study} | \text{Passed}) = P(\text{Pass} | \text{Study}) \times P(\text{Study}) / P(\text{Pass})$$

Each part of this formula has a specific role:

- $P(\text{Study})$ is how common studying is among students before we know anything about this particular individual.
- $P(\text{Pass} \mid \text{Study})$ is how probable it is to pass given that the student did study.
- $P(\text{Pass})$ is the overall probability of passing across all students, regardless of whether they studied.
- $P(\text{Study} \mid \text{Passed})$ is updated belief after taking the new evidence into account.

The result gives us information. Even though 90% of students who study pass the exam, 40% of students do not study at all, which means a large portion of those who pass did so without studying. Knowing that a student passed does increase our estimate that they studied, from 60% up to around 69%, but it does not confirm it.

Question 5:

	A	B	C	D	E	F	G
1	Q5 - Gaussian Kernel Regression at $x_0 = 3$, $h = 1$						
2							
3	x	y					
4	1	2					
5	2	4					
6	3	6					
7	4	8					
8							
9	$x_0 = 3$	$h = 1$					
10							
11	Kernel:	$K(u) = (1/\sqrt{2\pi}) \cdot \exp(-u^2/2)$ where $u = (x_0 - x_i) / h$					
12							
13	x_i	y_i	$(x_0 - x_i)$	$u = \text{diff}/h$	u^2	$K(u)$	$K(u) \cdot y_i$
14	1	2	2	2	4	0,053991	0,107982
15	2	4	1	1	1	0,241971	0,967883
16	3	6	0	0	0	0,398942	2,393654
17	4	8	-1	-1	1	0,241971	1,935766
18	Sum					0,936875	5,405285
19							
20	$\hat{Y}(x_0=3) = \Sigma K(u) \cdot y_i / \Sigma K(u) =$						5,7695
21							

Kernel regression is a non-parametric technique used to estimate the relationship between two variables, x and y , without assuming a specific mathematical form for the relationship (GeeksforGeeks, 2025).

The Gaussian kernel is a weighting function that assigns weights based on the distance between each observation and the query point

The predicted value at $x_0 = 3$ is calculated using the Nadaraya-Watson estimator:

$$\hat{Y}(x) = \frac{\sum K(u_i) y_i}{\sum K(u_i)}$$

This method calculates a weighted average of the observed y -values. Each observation contributes according to its kernel weight, and the final prediction is

obtained by dividing the weighted sum of the observations by the total weight (GeeksforGeeks, 2025).

Explanation of my results:

At $x = 3$, the observations closest to the query point receive the largest weights and therefore have the greatest impact on the prediction. Points that are further away contribute less because their weights are smaller. Since the data follows an almost perfectly linear pattern ($y = 2x$), the kernel regression estimate at $x = 3$ is very close to the actual value of $y = 6$. This shows that the Gaussian kernel regression model successfully captures the relationship between the variables and produces an accurate prediction.

Question 6:

	A	B	C	D	E
1	Q6 – Standardisation, Covariance Matrix, and Eigenvalues				
2					
3	Raw Data				
4		Feature A	Feature B	Feature C	
5	Sample 1	4	3	6	
6	Sample 2	10	2	4	
7	Sample 3	16	8	16	
8					
9	Mean ($\bar{x}$)	10,0000	4,3333	8,6667	
10	Std Dev σ (popula	4,8990	2,6247	5,2493	
11					
12	Standardised Data $z_{ij} = (x_{ij} - \bar{x}_j) / \sigma_j$				
13		Feature A	Feature B	Feature C	
14	Sample 1	-1,2247	-0,5080	-0,5080	
15	Sample 2	0,0000	-0,8890	-0,8890	
16	Sample 3	1,2247	1,3970	1,3970	
17					
18	Covariance Matrix $\text{Cov}(i,j) = (1/n) \cdot \sum (z_i)(z_j)$				
19		Feature A	Feature B	Feature C	
20	Feature A	1,0000	0,7777	0,7777	
21	Feature B	0,7777	1,0000	1,0000	
22	Feature C	0,7777	1,0000	1,0000	
23					
24					

6.1) Standardisation Formula:

Each value is standardised using: $z_{ij} = (x_{ij} - \bar{x}_j) / \sigma_j$

where $\bar{x}_j$ is the mean of feature j and σ_j is its population standard deviation (Lanhenke, 2021). The means and standard deviations for each feature are calculated in the Excel file.

Covariance Matrix:

The covariance matrix is calculated on the standardised data. Each entry $\text{Cov}(i, j)$ measures how much features i and j move together:

$$\text{Cov}(i, j) = (1/n) \times \sum z_i \cdot z_j$$

Because we are using standardised data, the diagonal entries equal 1.0 (the variance of a standardised variable is always 1) and the off-diagonal entries are the correlations between pairs of features. A covariance matrix computed on standardised data is equal to a correlation matrix (Lanhenke, 2021).

6.2)

24					
25	Eigenvalue Calculation – Q6.2				
26	Characteristic polynomial: $\det(C - \lambda I) = 0$				
27	$\lambda^3 - 3.0000\lambda^2 + 0.7903\lambda - 0.000000 = 0$				
28	where: $\text{trace}(C) = \sum C_{ii} = \text{coefficient of } \lambda^2$				
29					
30	Coefficient	Formula	Value		
31	Trace (coeff. of λ^2)	$C_{aa} + C_{bb} + C_{cc}$ (:	3,0000		
32	Sum of 2×2 minor	$(C_{aa} \cdot C_{bb} - C_{ab}^2$	0,7903		
33	Determinant (const	$\det(C)$	0,000000		
34					
35	Eigenvalues (roots of polynomial)				
36	PC	Eigenvalue	ariance Explained (	Cumulative (%)	
37	PC1	2,7082	90,27	90,27	
38	PC2	0,2918	9,73	100	
39	PC3	0,0000	0	100	
40					
41	Verification: $\lambda_1 + \lambda_2 + \lambda_3 = 3.0000 = \text{Trace}(C) = 3.0000 \checkmark$				
42					

Eigenvalues are found by solving the characteristic equation $\det(C - \lambda I) = 0$, where C is the 3×3 covariance matrix and I is the identity matrix (Miller, 2022). Expanding the determinant produces a cubic polynomial:

$$\lambda^3 - \text{trace}(C) \cdot \lambda^2 + (\text{sum of } 2 \times 2 \text{ principal minors}) \cdot \lambda - \det(C) = 0$$

The three coefficients (the trace, the sum of 2×2 minors, and the determinant) are all calculated in the Excel sheet using the standardised covariance matrix. Solving this cubic gives the three eigenvalues.

Explanation of my answer:

The three eigenvalues are shown in the Excel file alongside the percentage of total variance each explains. The largest eigenvalue (PC1) captures the dominant direction of variation in the data. A useful check is that the sum of all three eigenvalues must

equal the trace of the covariance matrix (which equals 3.0 for a standardised 3-feature dataset, since each diagonal entry is 1.0) (Miller, 2022). This verification is included in the Excel sheet.

If the first one or two eigenvalues explain the majority of the variance (say, 80% or more), we can project the data onto those components and discard the rest; reducing dimensionality while retaining most of the information. (Miller, 2022) This is the core purpose of PCA.

Reference List:

AIML.com (2022). *What is Bagging? How do you perform bagging and what are its advantages?* [online] AIML.com. Available at: <https://aiml.com/what-is-bagging/> [Accessed 15 Jun. 2026].

Alam, A. (2024). *Understanding the Difference Between Bagging and Random Forest.* [online] Medium. Available at: <https://medium.com/@aarishalam22/understanding-the-difference-between-bagging-and-random-forest-f0da72d7a3b2> [Accessed 15 Jun. 2026].

Balaji Nalawade (2024). *The Essential Guide to K-Fold Cross-Validation in Machine Learning.* [online] Medium. Available at: <https://medium.com/@bididudy/the-essential-guide-to-k-fold-cross-validation-in-machine-learning-2bcb58c50578> [Accessed 15 Jun. 2026].

Brownlee, J. (2019). *A Gentle Introduction to Bayes Theorem for Machine Learning.* [online] Machine Learning Mastery. Available at: <https://machinelearningmastery.com/bayes-theorem-for-machine-learning/> [Accessed 15 Jun. 2026].

GeeksforGeeks (2025). *Gaussian Kernel.* [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/machine-learning/gaussian-kernel/> [Accessed 15 Jun. 2026].

Gow, M. (2019). *Test Statistic Cheat Sheet: Z, T, F, and Chi-Squared.* [online] Medium. Available at: <https://medium.com/@marin892/test-statistic-cheat-sheet-z-t-f-and-chi-squared-58f4d9db5b99> [Accessed 15 Jun. 2026].

IBM (2021). *What Is a Decision Tree?* [online] IBM. Available at: <https://www.ibm.com/think/topics/decision-trees> [Accessed 15 Jun. 2026].

Kamble, V. (2022). *Fold Cross Validation - an overview* | *ScienceDirect Topics*. [online] [www.sciencedirect.com](https://www.sciencedirect.com/topics/computer-science/fold-cross-validation). Available at: <https://www.sciencedirect.com/topics/computer-science/fold-cross-validation> [Accessed 15 Jun. 2026].

Lanhenke, M. (2021). *Understanding the Covariance Matrix*. [online] Medium. Available at: <https://medium.com/data-science/understanding-the-covariance-matrix-92076554ea44> [Accessed 15 Jun. 2026].

Lee, F. (2025). *Logistic Regression*. [online] IBM.com. Available at: <https://www.ibm.com/think/topics/logistic-regression> [Accessed 15 Jun. 2026].

Miller, A. (2022). *Eigenvalues and Eigenvectors - an overview* | *ScienceDirect Topics*. [online] [www.sciencedirect.com](https://www.sciencedirect.com/topics/neuroscience/eigenvalues-and-eigenvectors). Available at: <https://www.sciencedirect.com/topics/neuroscience/eigenvalues-and-eigenvectors> [Accessed 15 Jun. 2026].

Nuttanan Wichitaksorn, Kang, Y. and Zhang, F. (2023). Random feature selection using random subspace logistic regression. *Expert Systems With Applications*, 217, pp.119535–119535. doi:<https://doi.org/10.1016/j.eswa.2023.119535>.

Ramchandani, P. (2018). *Random Forests and the Bias-Variance Tradeoff - TDS Archive - Medium*. [online] Medium. Available at: <https://medium.com/data-science/random-forests-and-the-bias-variance-tradeoff-3b77fee339b4> [Accessed 15 Jun. 2026].

Salman, H.A., Kalakech, A. and Steiti, A. (2024). Random Forest Algorithm Overview. *Babylonian Journal of Machine Learning*, [online] 2024, pp.69–79. doi:<https://doi.org/10.58496/bjml/2024/007>.

Sharma, N. (2015). *Hypothesis testing; z test, t-test, f-test*. [online] SlideShare. Available at: <https://www.slideshare.net/slideshow/hypothesis-testing-z-test-ttest-ftest/52262187> [Accessed 15 Jun. 2026].

Statsig.com. (2024). *Variance reduction techniques for accurate experiment results*. [online] Available at: <https://www.statsig.com/perspectives/variance-reduction-techniques> [Accessed 15 Jun. 2026].

